TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Single-cell and spatially resolved analysis of RNA vaccine fate after delivery			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		Bio-Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Vaccines	Medical Bio	Gene Pharmaceuticals
Concept of Proposed Technology	The proposed technology is a single-cell, spatially resolved analytical platform for mechanistic optimization of RNA vaccine performance. Once the RNA reaches the cytoplasm, it is translated directly by host ribosomes; however, the efficiency and uniformity of this process depend strongly on intracellular RNA fate, including spatial localization and degradation dynamics, which are not captured by conventional bulk assays. This platform integrates high-resolution single-cell imaging with quantitative gene-expression modeling to directly measure RNA's subcellular localization, degradation, and translation efficiency following delivery. Although RNA vaccine expression is inherently transient, immune protection depends on the magnitude, timing, and cellular context of antigen presentation rather than long-term expression. Therefore, optimizing transient expression dynamics at the single-cell level is critical for efficacy and dose efficiency. Inspired by recent super-resolution mapping of RNA spatial organization in cells, this project treats intracellular RNA localization as a controllable design variable. We will quantify where delivered vaccine mRNA resides at the nanoscale (endosomes, cytosol, ER-proximal regions, stress granules) and link spatial fate to translation output and cell-to-cell variability. The platform is designed as a generalizable analytical tool that complements existing experimental pipelines			

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	and accelerates rational RNA vaccine development.
Rationale for Joint R&D Project	Korean biotech SMEs developing RNA vaccines face major challenges in predicting intracellular RNA behavior, including delivery efficiency, degradation, spatial sequestration, and expression variability. These factors critically determine antigen production but remain largely invisible to bulk biochemical assays. A joint R&D project is required to integrate advanced single-cell and spatial imaging capabilities with industry-relevant RNA constructs and formulation strategies. My group contributes expertise in quantitative imaging and modeling of RNA lifecycle dynamics, while the Korean SME partner provides RNA vaccine designs, delivery platforms, and translational development constraints. This collaboration enables mechanistic insights that directly inform design decisions, reducing empirical trial-and-error and de-risking downstream animal studies. By focusing on early-stage mechanistic and model-based optimization, the project strengthens SME competitiveness and aligns with the innovation and commercialization goals supported by MSS.
Keywords	1. RNA vaccines 2. Single-cell RNA imaging 3. RNA stability 4. Spatial RNA localization 5. Expression heterogeneity

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	The objectives of this R&D project are to (1) establish a single-cell imaging framework to quantify intracellular fate and spatial localization of delivered vaccine mRNA at the single-cell level, (2) derive spatially targeted mRNA design rules that correlate subcellular RNA localization with expression levels, and (3) explore a "spatially programmed mRNA vaccine" strategy, in which vaccine designs are optimized not only for sequence-level stability and translation efficiency, but also for intracellular routing, away from degradation-prone compartments and toward translation-competent regions. Currently, we are capable of single-molecule and single-cell fluorescence microscopy (ref 1) and developing stochastic gene expression models (ref 2). For this project, we will develop RNA-specific labeling method for RNA vaccines and develop predictive models linking RNA sequence, subcellular localization to expression outcomes.			
R&D Target Specifications	See next section.			
R&D Plan				

Phase	Duration	Key activities	Milestones
1	0-12 months	Imaging assay development	Validate RNA imaging method
2	12-18 months	Data acquisition on existing RNA constructs	Quantitative RNA fate data for various RNA constructs
3	18-24 months	Model development	Predictive model
4	24-30 months	Model testing using new constructs	Demonstrate design improvements

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	We will develop and operate single-cell imaging and analysis pipelines. We will perform quantitative analysis of RNA localization, degradation, and translation dynamics upon entry. We will build and validate a model linking RNA sequence to spatial fate and gene expression. We will provide results and suggest RNA vaccine design principles to the biotech partners.
Expected R&R of Korean SME Partner	We expect the Korean SME partner to participate in experimental design and validation, provide multiple RNA vaccine constructs and delivery formulations, and synthesize new RNA vaccine sequences based on our findings.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project will deliver a validated single-cell and spatial RNA analysis platform that enables Korean SMEs to rationally optimize RNA vaccine performance. Expected outcomes include improved expression consistency, reduced required dose, faster design iteration, and lower development risk. By establishing spatially informed design principles for transient RNA expression, the project enhances Korea's competitiveness in RNA vaccines and broader RNA-based therapeutics, with strong commercialization and export potential.

6. References

(1) Troyer L, Wang YH, Shobhna, Kim S, Woo J, Tajkhorshid E, Kim S. Roles of the membrane-binding motif and the C-terminal domain of RNase E in localization and diffusion in *E. coli*. eLife. 2025 Nov 7; 14:RP105062.

(2) Kim S, Wang Y-H, Hassan A, Kim S. Re-defining how mRNA degradation is coordinated with transcription and translation in bacteria. bioRxiv 2024.04.18.588412.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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